# MACHINE LEARNING BASED LANDSLIDE SUSCEPTIBILITY MAPPING FRAMEWORK USING RANDOM FOREST AND EXPLAINABLE AI

# IN GILGIT DISTRICT (2017–2024)

A Final Year Project Thesis Submitted in Partial Fulfillment of the Requirements for the Degree of Bachelor of Science in Geography (Remote Sensing & GIS)


**SUBMITTED BY:**

Ishmal Imran…. (Roll No: 22011508-002)

Quratulain Khalid… (Roll No: 22011508-022)

Amna Mehdi……..(Roll No: 220150-024)

Sana Sabir……..(Roll No: 22011508-063)

**SUPERVISOR:**

Dr. Saira Munawar

Assistant Professor

Department of Geography, University of Gujrat


UNIVERSITY OF GUJRAT

A WORLD CLASS UNIVERSITY

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# DECLARATION

We hereby declare that the thesis titled “Machine Learning Based Landslide Susceptibility Mapping Framework Using Random Forest and Explainable AI in Gilgit District (2017–2024)” is our own original research work. It has not been submitted previously for a degree or any other qualification at this or any other institution. All sources of information used in this research have been duly acknowledged.

Signatures of Students:

1. Ishmal Imran ______________________________ Date: ____________

2. Qurat-ul-ain Khalid ________________________ Date: ____________

3. Amna Mehdi Khan _________________________ Date: ____________

4. Sana Sabir _________________________________ Date: ____________

Supervisor’s Signature:

Dr. Saira Munawar ______________________________ Date: ____________

Assistant Professor, Department of Geography, University of Gujrat

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# LIST OF ABBREVIATIONS

**AHP**… .......................................................................... Analytic Hierarchy Process

**ANN**… ......................................................................... Artificial Neural Networks

**ASTER GDEM** ................................... Advanced Spaceborne Thermal Emission and

Reflection Radiometer Global DEM

**AUC**… .................... .............................................................. Area Under the Curve

**CHIRP**.................... …..Climate Hazards Group InfraRed Precipitation with Station data

**CPE**........................................................................ China-Pakistan Economic Corridor

**DEM** ............................................................................. Digital Elevation Model

**DRR**… ............................................................................ Disaster Risk Reduction

**ESA**… ........................................................................... European Space Agency

**FoS** ................................................................................................. Factor of Safety

**FR**.................................................................................................... Frequency Ratio

**GBDMA** .............................................. Gilgit-Baltistan Disaster Management Authority

**GEE** ........................................................................................ Google Earth Engine

**GIS**.................................................................... Geographic Information System

**KKH…** ........................................................................................Karakoram Highway

**LCF**....................................................................................Landslide Conditioning Factor

**LR…** ........................................................................................... Logistic Regression

**LSM** ........................................................................ Landslide Susceptibility Mapping

**LULC**……………………………………………………Land Use / Land Cover

**ML** ................................................................................................. Machine Learning

**MKT** .........................................................................................Main Karakoram Thrust

**MMT**.............................................................................................Main Mantle Thrust

**NDVI**……………………………………..Normalized Difference Vegetation Index

**NHA**… ....................................................................... National Highway Authority

**OA**… ...............................................................................................Overall Accuracy

**PMD**… ............................................................Pakistan Meteorological Department

**RF** ...................................................................................................Random Forest

**ROC**………………………………………………..Receiver Operating Characteristic

**SHAP**… ...................................................................... Shapley Additive explanations

**SI**… ........................................................................................................ Statistical Index

**SPI**… ..............................................................................................Stream Power Index

**SVM**........................................................................................ Support Vector Machine

**TOL** ...........................................................................................................Tolerance

**TWI…**........................................................................Topographic Wetness Index

**UIB…**..........................................................................Upper Indus River Basin

**UNISDR** .......................United Nations International Strategy for Disaster Reduction

**VIF…**.......................................................................... Variance Inflation Factor

**WoE**............................................................................................ Weight of Evidence

**XAI…** ........................................................................Explainable Artificial Intelligence

**GPM**……………………………………………… Global Precipitation Measurement

**IMERG**……………………………….. Integrated Multi-satellite Retrievals for GPM

**DT**……………………………………………………………………..Decision Tree

**UTM** …………………………………………………..Universal Transverse Mercator

**CSV**………………………………………………………. Comma-Separated Values

**NDMA**………………………………………..National Disaster Management Authority

**GPM-IMERG** ….Global Precipitation Measurement Integrated Multi-satellite Retrievals for GPM

**DTM** ………………………………………………………………..Digital Terrain Model

**GEP**……………………………………………………………………... Google Earth Pro

**UTM Zone 43N** ,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,Universal Transverse Mercator Zone 43 North

# ABSTRACT

One of the major natural hazards in Gilgit District of Gilgit-Baltistan, mountainous area of Pakistan is the occurrence of landslides. The area has steep slopes, active tectonic movements, complex geology and more and more human activities that lead to slope instability. The purpose of this study was to create a framework for Landslide Susceptibility Mapping (LSM) for Gilgit District based on machine learning Random Forest (RF) model with the incorporation of Explainable Artificial Intelligence (XAI) methods.

The landslide inventory was developed based on historical landslide locations determined from Google Earth Pro and other disaster management materials. To create a balanced dataset for model training and validation, an equal number of non-landslide points were generated in ArcGIS. The 12 landslide conditioning factors chosen were the Elevation, Slope, Aspect, Curvature, Profile Curvature, Topographic Wetness Index (TWI), Stream Power Index (SPI), Land Use/Land Cover (LULC), Normalized Difference Vegetation Index (NDVI), Rainfall, Distance from Rivers and Distance from Roads. These factors were created using multi-source geospatial data such as DEM, Sentinel-2 imagery, WorldCover data from ESA, rainfall data from the Global Precipitation Measurement (GPM) IMERG products. Pre-processing included standardizing all data to a common spatial resolution and projection system.

No problem of collinearity was detected in the data through the multicollinearity test (Variance Inflation Factor (VIF) and Tolerance (TOL)) and all conditioning factors were appropriate for model development.

The Random Forest model was trained with an Overall Accuracy of 84.4%, Precision of 87.0%, Recall of 81.0%, F1-Score of 84.0% and Area Under the Curve of 0.94. The final susceptibility map was grouped into five susceptibility categories: Very Low, Low, Moderate, High and Very High susceptibility. SHapley Additive exPlanations (SHAP) were used to increase the interpretability of the model.

The analysis revealed the factors most influential were (in order): Slope, NDVI, Distance from Roads, Distance from Rivers, TWI and Elevation. The findings show that the primary factors that contribute to landslide susceptibility in the study area are steep slopes, vegetation cover, and road and river channels. The developed susceptibility map thus becomes a useful instrument to be used in disaster risk management, infrastructure planning and sustainable land use decision making in Gilgit District.

**Keywords:** Landslide Susceptibility Mapping, Random Forest, Explainable Artificial Intelligence, SHAP, Gilgit District, GIS, Remote Sensing, Machine Learning, GPM IMERG, Google Earth Pro

# CHAPTER 1

# INTRODUCTION

## 1.1 Background of the Study

A landslide is one of the most devastating natural disasters that occur in the mountains and can be a serious danger to lives, infrastructure and the environment. They are characterized by the downslope flow of rock, soil, debris or a mixture of these materials in response to gravity (Guzzetti et al., 2012; Reichenbach et al., 2018). Landslides are geological events that are part of the process by which the landscape evolves, but, when they threaten settlements, transport systems and economic activities, they become hazardous.

In many countries in the world, the occurrence of landslides has become more frequent and severe over the last few decades (Petley, 2012). Climate change and increasing human use of mountainous areas are mostly linked to this trend (Gariano & Guzzetti, 2016). Precipitation changes such as heavy rain and fast snowmelt may lead to changes in soil moisture and pore-water pressure, thus decreasing slope stability and causing landslides (Crozier, 2010; Iverson, 2000). Meanwhile, human activities, including deforestation, road construction, urban development and slope excavation, further disrupt natural conditions of the terrain, making it more susceptible to slope failure (Korup & Clague, 2009).

Northern Pakistan is one of the most prone areas of South Asia to landslides due to its complex geology and rugged topography (Shafique & van der Meijde, 2015). The region is located at the confluence of the Karakoram, Hindu Kush and the Himalayan mountain systems where the Indian plate is still colliding with the Eurasian plate. This tectonic movement has resulted in high rates of uplift, steep slopes, fractured rock formations and active faults, all of which make the slopes unstable (Searle et al., 2010).

The extreme relief, deep river valleys, steep slopes and complex geology of the area make Gilgit District a particularly vulnerable area for landslides in the state of Gilgit-Baltistan. The district experiences significant elevation changes and other factors such as seasonal rainfall, snowmelt processes and seismic activity further increase the probability of slope failures (Alvi et al., 2020). The district also has critical transportation corridors and developing towns that are becoming more vulnerable to landsliding.

One of the key aspects of the district is the Karakoram Highway (KKH) that could be considered a vital transportation link and also a vital part of the China–Pakistan Economic Corridor (CPEC). The natural slope conditions have been changed in many places of the district due to continuous expansion of roads, the construction of infrastructure and urban development (Saira & Miandad, 2023). Roads and other infrastructure are often damaged or cut off by landslides along these corridors, and economic losses are common (Hussain et al., 2021). Thus, the accurate and reliable Landslide Susceptibility Mapping (LSM) is crucial for disaster risk reduction and sustainable planning in region.

### 1.2 Problem Statement

The existing approaches to landslide susceptibility assessment are still limited in order to carry out effective DRR in Gilgit District. The Analytic Hierarchy Process (AHP) (Khaliq et al., 2023), Frequency Ratio (FR), Logistic Regression (LR) and Weight of Evidence (WoE) are commonly used heuristic and statistical methods in previous studies conducted in northern Pakistan. These techniques have helped to assess hazards, but they also rely on expert judgement or simple statistical assumptions about the relationships between hazard triggering which may not capture the complexity of the relationships controlling landslide occurrence (Reichenbach et al., 2018).

Topography, hydrology, vegetation, land use, and rainfall are all factors that play a role in the influence of mountain environments in Gilgit (Basharat et al., 2021). Therefore, the use of conventional mapping approaches can lead to uncertainties and misclassification, which makes susceptibility maps less useful for planning and risk management purposes.

Machine Learning (ML) has seen significant developments in recent years and this has opened new possibilities to enhance landslide susceptibility mapping. Of these techniques, Random Forest (RF) has shown to be a robust model because it can model complex non-linear relationships, can cope with mixed datasets and can minimize overfitting issues (Breiman, 2001; Basharat et al., 2021). Machine learning models have been proven to be predictive, but are also known to have poor interpretability in that the decision-making logic isn't necessarily obvious to decision makers (Lundberg & Lee, 2017).

Another significant research void is related to recent changes in Gilgit District with regard to the environment and human induced changes. Rainfall pattern changes, progression of infrastructure and rise in human activities on mountain slopes were observed during the period 2017 to 2024 in the region (Moazzam et al., 2022). The knowledge of how these factors affects the landslide susceptibility is useful for the construction of effective landslide risk reduction strategies.

In this context, this study aims to tackle these challenges by combining two powerful machine learning techniques: Random Forest (RF) and SHapley Additive exPlanations (SHAP), to create an interpretable and accurate landslide susceptibility map for Gilgit District.

The use of RF and SHAP can enhance prediction accuracy and reveal the contribution of various environment factors on landslides, which helps to make better decision for disaster management and land-use planning.

## 1.3 Objectives of the Study

The main aim of this study is to design and validate a data driven Landslide Susceptibility Mapping (LSM) framework for Gilgit District by machine learning and explainable AI (XAI) approach in the context of GIS and remote sensing.  The specific objectives are:

- To develop a dependable dataset for the landslide inventory for Gilgit District for the period 2017-2024 using the imagery from Google Earth, disaster reported or predocumented landslide events.
- To gather, preprocess, standardize and map 12 landslide conditioning factors (topographic, hydrological, environmental and anthropogenic), such as Elevation, Slope, Aspect, Curvature, etc.  Profile Curvature, TWI, SPI, NDVI, LULC, Rainfall, Distance from Rivers, and Distance from Roads.
- To add multicollinearity analysis (variance Inflation Factor or Tolerance) to identify any possible relationships between the conditioning factors.
- To build and apply Random Forest (RF) model to predict landslide susceptibility and compare the results.To build and use Random Forest (RF) model for landslide susceptibility prediction and compare the results, using a training-testing dataset split of 70:30 respectively.
- To test the accuracy, precision, recall, f1 score, confusion matrix and roc-auc of the model.
- To use SHapley Additive exPlanations (SHAP) to interpret model predictions, and recognise the relative impact of the conditioning factors on landslide occurrence.

- To develop a Landslide Susceptibility Map of Gilgit District, and to classify the area into different susceptibility zones for Disaster Risk Reduction & Land use Planning.

## 1.4 Research Questions

This thesis provides empirical, structural, and computational answers to the following central research questions:

- What is the exact spatial distribution and historical characteristics of the landslide events recorded in Gilgit District through using multi-temporal remote sensing observations during 2017-2024?
- How much are the 12 selected geo-environmental conditioning factors statically independent and what is the advantage of removing multicollinearity in order to increase the stability of the prediction of the models based on data?
- What is the ability of an ensemble Random Forest architecture (classifier ensemble) to classify the terrain of Gilgit into different susceptibility levels and what are its exact validation bounds by using ROC-AUC matrices?
- What are the specific scenarios for the global ranking configurations and local marginal thresholds of the conditioning factors as looked at from a cooperative game-theory perspective (Lundberg & Lee, 2017)?
- What are the implications for a more effective way of using XAI to provide regional planning and disaster mitigation decision makers with a tool for their decision making that is both actionable and legally accountable that can replace the “black-box” machine learning susceptibility map?

## 1.5 Significance of the Study

### 1.5.1 Theoretical and Academic Significance

Academic contribution of this study includes engineering geology, geomorphology and landslide susceptibility modelling which are integrated in GIS and Remote Sensing, while using Machine Learning and Explainable Artificial Intelligence (XAI) techniques in Pakistan. In recent studies in geohazard science, the main focus has been on the prediction accuracy of the model, however this study takes a step further by using a combination of two methods (Random Forest (RF) and SHAP) to not only improve the prediction accuracy of the model but also to make it interpretable (Kunwar et al., 2025; Inan & Rahman, 2023).

The study presents empirical evidence of the effect of topographic, hydrological, environmental and anthropogenic factors on landslides occurrence in the study area, Gilgit District. It also exemplifies a repeatable methodology which can be customized for landslide susceptibility mapping in other mountainous regions of Pakistan and the entire Hindu Kush–Karakoram–Himalayan region. Furthermore, the incorporation of conditioning information through SHAP enables an understanding of how conditioning factors affect model predictions and can serve to connect the science of modelling to understanding the science and the model's predictions.

### 1.5.2 Practical, Institutional, and Engineering Significance

In terms of applications, the landslide susceptibility map produced in this study can help in the disaster management, planning of infrastructure and decision making in land use in Gilgit District. High and very high susceptibility areas are highly useful for the planning agencies at the national level (National Disaster Management Authority (NDMA), the National Highway Authority (NHA), Ministry of Climate Change), also for the planning agencies at local level (GBDMA), and for the Gilgit-Baltistan Disaster Management Authority (GBDMA) (see 2024).

This can help engineers and planners recognize areas of transportation routes, settlements, and river valleys which are at risk of slope failure, in which stabilization, monitoring, and mitigation measures should be considered. In addition, using SHAP will help to understand the relative impact of each conditioning factor, which can inform better and evidence-based decision making. Finally, the results of this research can be helpful in the field of disaster risk reduction, sustainable infrastructure development, and building community's resilience against landslide hazards in Gilgit District.

# CHAPTER 2

# LITERATURE REVIEW

## 2.1 Global and Regional Landslide Vulnerability Dynamics

Worldwide, landslides are one of the most important natural hazards, and can trigger many fatalities, economic losses and environmental impacts annually (Petley, 2012). The United Nations International Strategy for Disaster Reduction (UNISDR, 2015) states that the geology and steepness of mountain areas increase the risk of landslides for mountainous regions like the Andes, the Alps and the Himalayan-Karakoram region. According to Reichenbach et al. (2018), climate change and the increased human activities are the main causes affecting the occurrence of landslides all over the world.

Recent studies have demonstrated that the risk of slope failure caused by extreme weather events, such as heavy rainfall, cloudbursts and glacier melting is related to the increase in soil moisture and pore-water pressure inside slopes (Gariano & Guzzetti, 2016). Unstable slopes, particularly in mountainous areas are important in these conditions. This has led to more landslides and debris flows in many areas of the Hindu Kush – Himalayan region (Bajracharya et al., 2018).

South Asia is considered to be one of the most landslide prone regions of the world at the regional scale. Many countries like Pakistan, Nepal, Bhutan and southwestern China are prone to landsliding in response to the active tectonic activity, topography, and seasonal rainfall (Korup & Clague, 2009). The northern mountainous region is especially vulnerable in Pakistan due to the presence of river networks, extensive fault systems and rough topography in this area. The development of infrastructure, particularly along the Karakoram Highway (KKH), has led to an increase in slope instability and landslide events in the area of Gilgit-Baltistan, which is one of the most vulnerable regions (Hussain et al., 2021; Saira & Miandad, 2023).

Additionally, other climatic changes and the rise in climatic variability have exacerbated the landslide hazard in many regions of the country (Moazzam et al., 2022).

## 2.2 Theoretical Framework of Slope Instability and Mass Wasting

Mass wasting" is defined as the down slope movement of soils, rocks, regolith and other earth materials under the influence of gravity (Varnes, 1978; Hungr et al, 2014). Several geomorphological processes including rockfalls, debris flows, earthflows, and landslides are included. Of these, landslides are one of the most harmful natural hazards because of their direct effects on human life, infrastructure and the environment (Petley, 2012).

The stability of a slope is normally expressed in terms of the Factor of Safety (FoS), which can be calculated as the ratio of resisting forces on a slope to the driving forces. Mathematically, it can be expressed as:

$$FoS = \tau_n / \tau$$

where $\tau_n$ represents The shear strength of the slope material is the shear strength of the slope material and $\tau$ is the driving shear stress imposed on the slope (Bishop, 1955). The shear strength is primarily dependent on the cohesion of soil or rock, the internal friction angle of the soil or rock, and the effective normal stress, while the driving stress is dependent mainly on the gravity and the slope gradient. A slope is said to be stable if the FoS is greater than 1.0 and unstable when the FoS falls below 1.0.

The factors that affect the slope stability are geological, climatic and tectonic factors in mountainous regions, especially in the Karakoram and Hindu Kush ranges (Searle et al., 2010; Shafique & van der Meijde, 2015). Some of the most common causes of landslides include heavy rainfall, fast snowmelt and earthquakes. These processes generate pore-water pressure in the slope materials that leads to the reduction of effective normal stress and, thus, the shear strength of soils and rocks (Iverson, 2000; Crozier, 2010). This makes the slopes more vulnerable to failure.

Hoek and Bray (1981) have previously investigated the geotechnical characteristics and the effect of weathering on shear strength parameters of materials, and found that both the cohesion (c) and internal friction angle ($\varphi$) are markedly influenced by the weathering conditions, material composition and grain-size distribution. The rock types in Gilgit District are mainly highly fractured slate, schist and gneiss, which are generally susceptible to structural weakness and weathering. (Alvi et al., 2020; Shafique & van der Meijde, 2015).

Further, rainfall infiltration and snow melt can lead to an increase of moisture within the slope material and further decrease its stability and increase the likelihood of rotation and translation landslides. The knowledge of these slope stability mechanisms can also be used as the theoretical basis for landslide susceptibility mapping because the environmental and topographic factors that influence the probability of slope stability failure in mountainous areas.

## 2.3 Geological and Climatic Setting of Northern Pakistan

Northern Pakistan lies in one of the world's most seismically active regions, where the Karakoram, Hindu Kush and Himalayan mountain ranges meet (Searle et al., 2010). This is the tectonic environment of a continuing collision between the Indian and the Eurasian plates, where the Indian plate is still continuing to move northward at a rate of about 40–50 mm per year (Jade et al., 2014).

This tectonic activity has created huge folding, faulting and crustal deformation which has resulted in the formation of major geological structures like Main Karakoram Thrust (MKT) and Main Mantle Thrust (MMT). Most of the rocks present are highly weathered schist, slate, gneiss and fractured limestone, which are prone to slope failure (Shafique & van der Meijde, 2015; Alvi et al., 2020).

Climatically, the District of Gilgit is defined as a high altitude cold desert with steep slopes, deep valleys of the region and huge height difference over short distances (Alvi et al., 2020). Weathering processes are important factors in reducing the strength of rock masses and enhancing the instability of slopes, especially freeze–thaw action (Matsuoka & Murton, 2008).

Further, the seasonal rains brought by the western disturbances and snow melt also result in higher runoff and soil saturation and subsequently lead to erosion. Further frequent seismic activity causes already fractured rock formations to break even more, leading to a higher risk of landslides or other mass movement processes (Shafique & van der Meijde, 2015). Northern Pakistan (more specifically in the Gilgit District) is very prone to landslide hazards due to the factors of active tectonics, complex geology, steep topography and climatic influences.

## 2.4 Human Activities and Slope Instability

Along with natural factors, human activities have also been found to be a major factor in making the lands in the Gilgit District more susceptible to landslides (Saira & Miandad, 2023). The natural stability of many mountain slopes has been changed due to fast development of infrastructure, especially under the China-Pakistan Economic Corridor (CPEC).

Road construction usually includes blasting, excavation and slope cutting operations, which affect the natural equilibrium of the slope material and raise the risk of slope failure (Iqbal et al., 2020). Road construction can further impair slope stability by removing their toe support and induce rotational and/or translational landslides.

More and more people have also brought more stress to the sensitive mountain ecosystems. Alluvial fans, debris-covered slopes and other potentially unstable areas have been the hosts of many settlements.

Besides, deforestation for fuelwood collection and agricultural expansion leads to loss of vegetation cover and root reinforcement, and thus to increased slope susceptibility to erosion and slope failure (Piacentini et al., 2012). Excessive irrigation can also lead to higher soil moisture and less slope stability caused by unplanned irrigation practices.

Landslide hazard can be greatly exacerbated by the effects of both human activities and environmental conditions. Heavy rainfall periods are more likely to cause disturbed slopes to experience debris flows, rockfalls, and other mass movements (Korup & Clague, 2009). Moreover, the ongoing development of transportation systems has also changed the types of topography and the spatial distribution of landslides in the district (Hussain et al., 2021; Saira & Miandad, 2023). So anthropogenic factors need to be taken into account along with natural factors in determining landslide susceptibility in mountainous region.

## 2.5 Evolution of Landslide Susceptibility Mapping (LSM) Methodologies

In the last decades, with the advent of new analytical methods and computational tools, Landslide Susceptibility Mapping (LSM) has undergone significant development. LSM methods can broadly be classified into three categories: heuristic, statistical and machine learning methods (Reichenbach et al., 2018).

The early studies on landslide susceptibility focused primarily on heuristic methods, which were based on the knowledge and field experience of the experts. The Analytic Hierarchy Process (AHP) has been one of the most used methods that assigns weight for a conditioning factors based on the expert judgment (Saaty, 1980). These methods are comparatively simple and straightforward, but they are subjective and have limited effectiveness for large areas of study (Yalcin et al., 2011).

Later, statistical approaches were developed to assess landslide susceptibility to improve objectivity. The use of some techniques to find the relation between landslide occurrences and environmental factors including Frequency Ratio (FR), Weight of Evidence (WoE), Statistical Index (SI), and Logistic Regression (LR) gained popularity in the last ten years (Lee & Talib, 2005; Ozdemir & Altural, 2013). These approaches offered a more quantitative approach to hazard assessment and less reliance on expert judgment.

While useful, traditional statistical models tend to make simplifying assumptions about the relationship between the occurrence of landslides and predictor variables, such as a simple relationship or a linear relationship. Landslides in complex mountainous areas like Northern Pakistan are influenced by many factors that can have non-linear relationships with each other (Basharat et al., 2021). Conventional statistical methods can therefore not fully represent the complexity of landslide processes and can also lead to lower predictive performance (Reichenbach et al., 2018).

These constraints have spurred the use of machine learning methods that have the potential to capture complex relationships between environmental factors more accurately. Because of this, in recent research on landslide susceptibility, machine learning techniques are gaining popularity and are being considered as a very useful tool for landslide hazard prediction.

### 2.6 Application of Random Forest (RF) in Landslide Susceptibility Mapping

In recent years, some studies have started using Machine Learning (ML) techniques for landslide susceptibility mapping and hazard assessment (Reichenbach et al., 2018; Merghadi et al., 2020) to address the limitations of traditional statistical methods.

Machine learning algorithms can detect the complicated relationship between environmental variables and demonstrate better predictive power than numerous traditional algorithms. Artificial Neural Networks (ANN), Support Vector Machines (SVM), Decision Trees (DT) and Random Forest (RF) are commonly used algorithms (Chen et al., 2017).

Random Forest (RF) algorithm, proposed by Breiman (2001), has emerged as one of the most popular methods in landslide susceptibility studies among these methods. Random Forest is an ensemble learning method in which multiple decision trees are created in the training process and their results are aggregated to make prediction. This gives a better prediction, decreases over-fitting (Breiman, 2001).

Random Forest is a highly suitable algorithm for landslide susceptibility mapping (Khaliq et al., 2023; Basharat et al., 2021) due to several reasons:

• **Ability to Handle Different Data Types:** Both continuous and categorical variables like elevation, rainfall, slope, land use/land cover (LULC) and aspect classes (Belgiu & Drăguţ, 2016) can be processed well using RF.

• **Resistance to Overfitting:** RF is able to combine the results of several decision trees, which helps the model to be more stable and less sensitive to noise in the data set (Breiman, 2001).

• **Performance with Correlated Variables:** RF can predict well even moderate level of correlation exists in some predictor variables (Merghadi et al., 2020).

Due to these benefits, Random Forest is successfully used in various landslide susceptibility studies in mountainous areas, such as northern Pakistan (Hussain et al., 2021; Khaliq et al., 2023). Hence, RF was chosen as a first choice modelling technique for this study.

## 2.7 Explainable Artificial Intelligence (XAI) and SHAP in Landslide Studies

Many machine learning models are hard to understand, but they can still be highly accurate in predicting outcomes. This can be called the “black box” problem, as the model's internal decision-making process is not easily understood (Lipton, 2018). When it comes to applications such as hazard assessment and disaster management, it is not only important that predictions are accurate but also that they are timely. In addition, decision-makers must be aware of which environmental factors are responsible for enhancing or reducing landslide susceptibility (Doshi-Velez & Kim, 2017).

Researchers have been turning to Explainable Artificial Intelligence (XAI) techniques (Arrieta et al., 2020) to enhance the transparency of the models. XAI methods provide insights into how machine learning models make predictions, and which predictor variables are most important.

The most popular XAI methods are based on SHapley Additive exPlanations (SHAP) of Lundberg and Lee (2017). SHAP is a game-theoretic approach that quantifies the impact of each predictor variable on the model prediction. The resulting value is called the Shapley Value, and it shows if there is an increase or decrease in the predicted susceptibility.

One of the strengths of SHAP is its capacity to give interpretation at both a global and local level. The global interpretation determines the global significance of the conditioning factors over all the data and the local interpretation explains the influence of conditioning factors at particular point(s) (Lundberg & Lee, 2017; Inan & Rahman, 2023). It enables researchers to gain insight into the influence of the environment on landslide occurrence.

Recent studies have shown that the combination of SHAP and machine learning models enhance the transparency and trustworthiness of landslide susceptibility mapping (Kunwar et al., 2025). Hence, in this study, the predictions of the Random Forest model explained using SHAP and the most influential factors identified for landslide occurrence in the Gilgit District were towards this end.

### 2.8 Research Gap

Based on the literature review, it is found that landslide susceptibility mapping is an interesting topic in northern Pakistan and other mountainous areas of the country. However, most studies have been centered on the prediction accuracy which relies either on traditional statistical methods or machine learning algorithms (Alvi et al., 2020; Hussain et al., 2021; Basharat et al., 2021). Less focus has been placed on model interpretation and understanding the individual contributions of the conditioning factors.

Furthermore, the majority of the previous studies conducted in the region of Gilgit-Baltistan were limited to particular road segments or small-scale areas and used the small size of the data sets. The changing environment, infrastructure construction and growing urban areas have impacted the slopes in the region, necessitating the need to update the susceptibility assessment (Saira & Miandad, 2023; Moazzam et al., 2022).

Moreover, there are limited studies in the Gilgit District that have incorporated machine learning with Explainable Artificial Intelligence (XAI) techniques like SHAP for landslide susceptibility mapping.

Hence, there is a need for an interpretable, data-driven landslide susceptibility framework which can be used to predict locations of landslide prone areas and elucidate reasons for the predictions. To tackle this issue, this study combines the Random Forest modelling with SHAP-based interpretation by utilizing geospatial datasets from 2017 to 2024. The suggested framework proposes to create susceptibility mapping as well as a better understanding of the environmental factors responsible for landslide occurrences in the Gilgit District.

# CHAPTER 3

# STUDY AREA AND METHODOLOGY

## 3.1 Geographic and Socio-Environmental Context of the Study Area

Gilgit District is part of the Northern Province of Gilgit-Baltistan, in northern Pakistan, and the economic and administrative hub of the region. The district is located in the latitude of 35°40′N to 36°40′N and longitude 73°40′E to 74°40′E. It is surrounded by Hunza district in north/north-east, Skardu and Shigar in the east, Ghizer in the west and Diamer in the south. The total area of the district is 4,208 km² (GBDMA, 2024).

Rugged mountainous area, deep valleys and steep slopes define the district. The settlements are mostly located along the valleys of rivers, alluvial fans and river terraces of Gilgit and Hunza (Alvi et al. 2020). Between 2017 and 2024, population growth, urbanization and infrastructure construction activities expanded in mountainous regions. Thus, there has been a growing exposure of many settlements, roads, schools, and public buildings to the landslide hazard (Hussain et al., 2021).

## 3.2 Tectonic and Structural Framework

The Gilgit District is situated in the active collision regime of the Indian and the Eurasian tectonic plates. The current collision of these plates is estimated at roughly 40-50 mm/yr, leading to a region of extensive faulting, folding and crustal deformation (Jade et al., 2014; Searle et al., 2010).

The area is affected by several large scale tectonic structures, of which the Main Karakoram Thrust (MKT) is a significant one as it is important in controlling the regional geology and seismic activity (Shafique & van der Meijde, 2015). Ongoing tectonic movements have created very fractured and weathered rock masses, causing a high risk of instability on many slopes. Other landform destabilising factors include earthquakes and tectonic movements which weaken the slope materials and decrease their stability (Alvi et al., 2020).

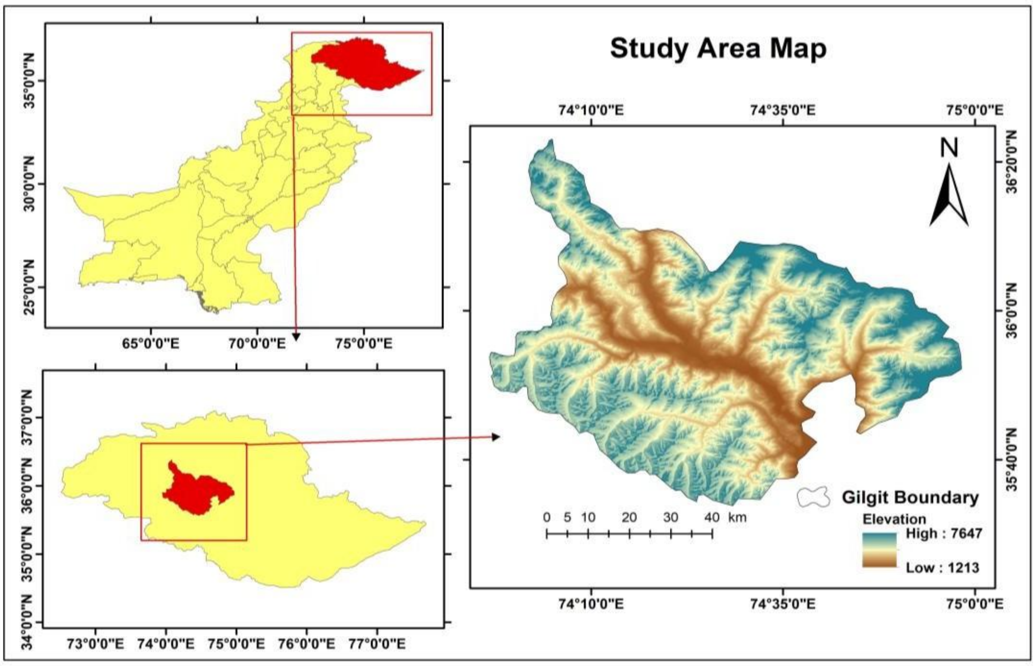

**Figure 1 : Study Area Map of Gilgit District, Gilgit-Baltistan, Pakistan**

### 3.3 Topography and Hydrological Characteristics

The topography of Gilgit District is very diverse ranging from around 1500 m in valleys to over 5500 m in the surrounding mountains (Alvi et al., 2020). Steep slopes, narrow valleys and rugged mountainous areas predominate. In many places, gradients are greater than 35° and slope failures and mass movement processes are more likely (Shafique & van der Meijde, 2015).

The district forms part of the Upper Indus Basin and is mainly drained by the Gilgit and Hunza Rivers. These rivers and their tributaries have formed deeply incised valleys by continuous erosion. In erosion along a valley's sides, the slope bases are often undercut, which decreases slope stability and increases the likelihood of landslides. Besides, many seasonal streams and mountain channels act as channels of debris flows and flash floods during heavy rainfall events (Moazzam et al., 2022).

### 3.4 Climate and Weathering Processes

The climate of Gilgit District is cold arid and semi arid mountain. Annual precipitation is a strong function of elevation. The general trend is that higher elevations in the valley receive more rainfall as compared to lower elevations, where the rainfall is generally low, especially in winter, when precipitation occurs mostly in the form of snowfall (PMD, 2024).

Seasonal weather changes, rainfall events and snow melt are significant factors affecting landslide activity in the area. Slope instability resulting from changes in rainfall intensities and localized extreme weather events occurred in a few areas during the study period (2017-2024) (Moazzam et al., 2022).

One of the major geomorphic processes is freeze-thaw weathering in this district. During cold weather, water seeps into cracks and joints in rock masses, turning to ice in freezing temperatures which eventually loosens the structure of the rock. Repeated freeze-thaw cycles also enhance rock fragmentation and will help susceptibility to slope instability and landsliding (Matsuoka & Murton, 2008).

### 3.5 Systematic Research Methodology Architecture

The methodology of this research is systematic which is conducted to prepare a landslide susceptibility map for the Gilgit District by incorporating GIS, Remote Sensing, Random Forest (RF) and Explainable Artificial Intelligence (XAI). The workflow takes place over five main phases, starting with preparation of landslide inventories and ending with susceptibility mapping and model interpretation.
The overall methodology includes:

- Preparation of the Landslide Inventory Dataset (2017–2024).
- Twelve landslide conditioning factors were collected and preprocessed.
- Variance Inflation Factor (VIF) and Tolerance (TOL) were used to check for multicollinearity.
- The model was trained and validated with a training-testing split of 70% and 30% respectively.The model was trained and validated with 70% training and 30% testing.
- Interpretation of models with Shapley Additive explanations (SHAP).
- Final Landslide Susceptibility Map (LSM) Creation with Factor Importance Weights.

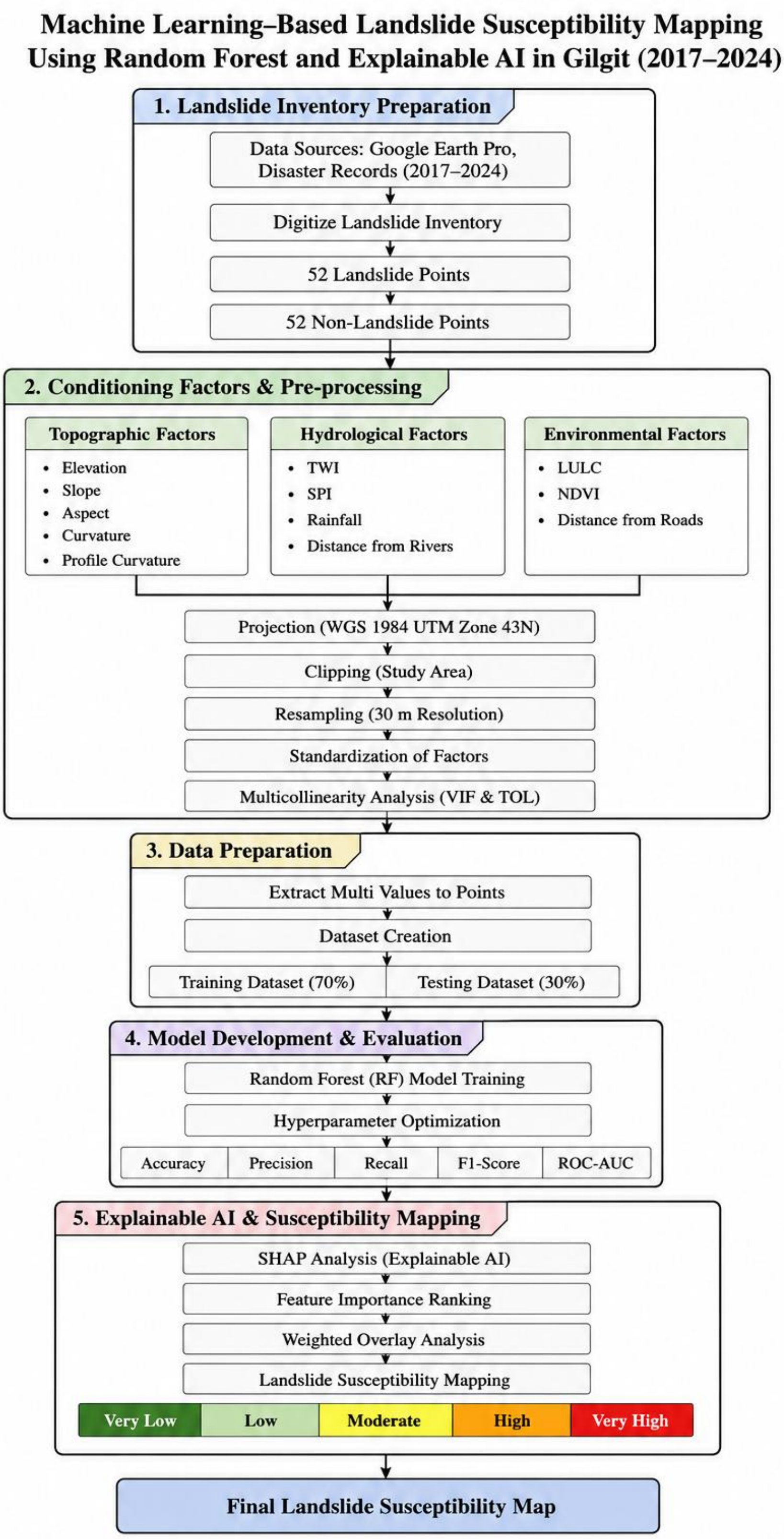


**Figure 2: Methodological Framework for Landslide Susceptibility Mapping Using Random Forest and Explainable AI**

### 3.6 Landslide Inventory Preparation

A trustworthy landslide inventory is necessary for training and testing machine learning models for landslide susceptibility mapping. A landslide inventory has been created for Gilgit District from 2017 to 2024 in this study. The historical locations of landslides were identified using visual interpretation of high-resolution satellite images of the area available in Google Earth and by referring to historical disaster reports received from relevant agencies.

The occurrence and location of landslide events in the study area were confirmed using satellite imagery and history. There were 104 inventory points, 52 of which were landslide and 52 were non-landslide. All the points were selected with the knowledge that landslides had occurred or were reported at the landslide points and no evidence of landslide activity was found at the non-landslide points, which were selected from relatively stable areas.

To develop a balanced dataset for supervised classification, an equal number of landslide and non-landslide samples were used. All of the values of the conditioning factors were calculated at each inventory location, and then combined into a single database for use in model development. The entire data set was split in the ratio of 70:30 with the training set being the majority and testing set being the minority. The samples were divided into training and independent testing sets with 70% for model training and the remaining 30% for independent model validation and performance evaluation.

### 3.7 Extraction and Standardization of Landslide Conditioning Factors (LCFs)

Topographic, hydrologic, environmental and anthropogenic conditions all play a role in landslide occurrence. Thus, 12 landslide conditioning factors have been chosen to be used in this study, which were considered to be relevant in previous landslide susceptibility studies and were available for the Gilgit District.

The factors selected were Elevation, Slope, Aspect, Curvature, Profile Curvature, Stream Power Index (SPI), Distance from Rivers, Distance from Roads, Topographic Wetness Index (TWI) and Land Use/Land Cover (LULC), Normalized Difference Vegetation Index (NDVI) and Rainfall.

The primary dataset was a Digital Elevation Model (DEM) used to calculate topographic parameters and hydrological parameters such as Elevation, Slope, Aspect, Curvature, Profile Curvature, TWI and SPI. Terrain and drainage-based layers were created using hydrological tools provided in ArcGIS.

The ESA WorldCover data set provided the land use/land cover information and the NDVI was derived from Sentinel-2 satellite data using the basic vegetation index equation. The Global Precipitation Measurement (GPM IMERG) data was used for the rainfall information needed for this report, which provides high-resolution precipitation data for use in regional environmental analysis. The Euclidean Distance tool in ArcGIS was used to create Distance from Rivers and Distance from Roads.

All datasets were projected to the WGS 1984 UTM Zone 43N coordinate system, and then resampled to 30 m to ensure consistency across all the datasets. Raster layers were clipped to the extent of the Gilgit District and prepared for further analysis. The Extract Multi Values to Points tool in ArcGIS was used to extract values for all 12 conditioning factors at every point in the inventory for each of the landslides and non-landslides.

The values extracted were organized into a single attribute table and then exported as a CSV dataset to conduct statistical analysis, test for multicollinearity, exploratory modelling using the Random Forest and Explainable Artificial Intelligence (SHAP) analysis.

### 3.7.1 Topographic Factors

- **Elevation:** The Digital Elevation Model (DEM) was used to generate the elevation, with a spatial resolution of 30 m. Elevation is an indirect variable that affects regional climatic conditions, the distribution of vegetation, the intensity of weathering, and snowmelt processes, and thus, it plays an important role in landslide susceptibility assessment (Reichenbach et al., 2018).

- **Slope:** The Slope tool in ArcGIS was used to create slope from the DEM. It is the steepest possible elevation change per pixel, and one of the most significant controls for landslide occurrence, since increased gravitational shear stress on slopes (Iverson, 2000) will cause increased elevations change.

- **Aspect:** Aspect is a derived variable from the DEM which describes the direction a slope is facing. It has indirect effects on the stability of slopes, because it can influence solar radiation, moisture content, vegetation growth and freeze–thaw. (Reichenbach et al., 2018).

- **Curvature:** The curvature of the terrain surface indicates the overall shape of the terrain and is used to describe if an area is concave, convex or close to flat. Concave surfaces will tend to collect water and loose soil, which will lead to saturated soils and landslide potential, while convex surfaces will tend to encourage water and soil to flow away. For this reason, curvature is a significant slope stability and hydrological surface process controlling factor.

- **Profile Curvature:** The curvature of the terrain parallel to the slope direction is called profile curvature. It can either increase or decrease the velocity of runoff and/or downslope movement of material which in turn can impact erosion and slope processes (Brazier, 2019).

### 3.7.2 Hydrological Conditioning Factors

- **Topographic Wetness Index (TWI):** Topographic Wetness Index (TWI) is a hydrological parameter that is used to quantify the distribution of soil moisture and potential water accumulation on the landscape (Beven & Kirkby, 1979). Regions with higher TWI values will have more moisture, and thus will be more prone to slope instability. TWI was derived from the DEM using the equation: $TWI = \ln(\alpha / \tan\beta)$

  Where $\alpha$ represents the specific catchment area and $\beta$ represents the slope angle.

- **Stream Power Index (SPI):** Erosive power of surface runoff and flowing water over the terrain is measured by Stream Power Index (Brazier, 2019). High SPI values tend to lead to more erosion and slope weakening in areas. SPI was computed based on the following equation:

  $SPI = \alpha \times \tan\beta$

  Where $\alpha$ is the flow accumulation and $\beta$ is the slope angle.

- **Distance from River Networks:** The Euclidean Distance tool was applied in ArcGIS to create a distance from river networks surface. A drainage network was separated from the river channels and distances between each pixel were calculated. Slopes close to rivers tend to be more prone to landslides due to continuous toe erosion and bank undercut which leads to the loss of stability of the slope (Hussain et al., 2021).

### 3.7.3 Environmental, Meteorological, and Anthropogenic Factors

- **Land Use / Land Cover (LULC):** The ESA WorldCover dataset, which has a spatial resolution of 10 m, was used for the Land Use/Land Cover data. The data were reclassified into major land-use/cover classes related to the study area. The effects of various land-cover types on vegetation cover, surface runoff and soil protection affect landslide occurrence. The slope failure susceptibility is typically greater on barren land and areas with low vegetation cover compared to higher vegetation cover areas (Khaliq et al., 2023).

- **Normalized Difference Vegetation Index (NDVI):** Sentinel-2 images were used to create NDVI using the standard vegetation index equation. The vegetation density and root cohesion were represented by NDVI. Generally, areas with higher NDVI values have a dense vegetation cover, which contributes to increased soil stability on the slope by improving soil integrity and reducing surface erosion (Gorelick et al., 2017; Khaliq et al., 2023).

- **Precipitation:** IMERG (Global Precipitation Measurement Integrated Multi-satellite Retrievals for GPM) data was used to compute precipitation. Among the most significant triggering factors for landslides mentioned by Crozier (2010) and PMD (2024), rainfall stands out as one of the most important because it raises the pore water pressure and soil moisture content, which leads to a reduction in slope stability and increases the risk of slope failure.

- **Distance from Roads:** The Euclidean Distance tool in ArcGIS was used to create Distance from Roads based on Karakoram Highway Road Network. Slopes near roads are more likely to suffer from landslides because of road construction, blasting, excavation, and slope cutting (Saira & Miandad, 2023; Hussain et al., 2021).

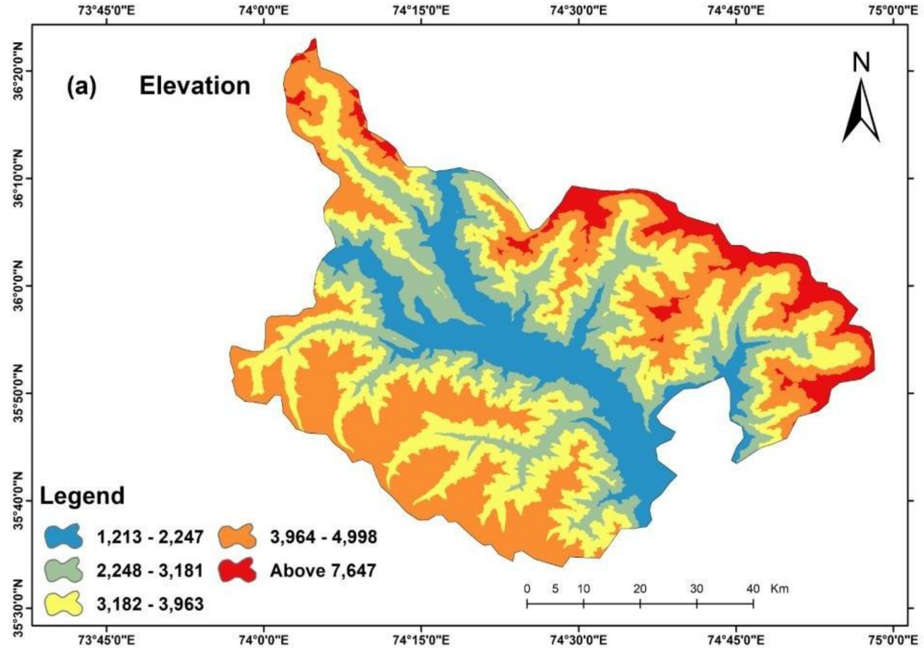


**Figure 3 : a) Elevation**

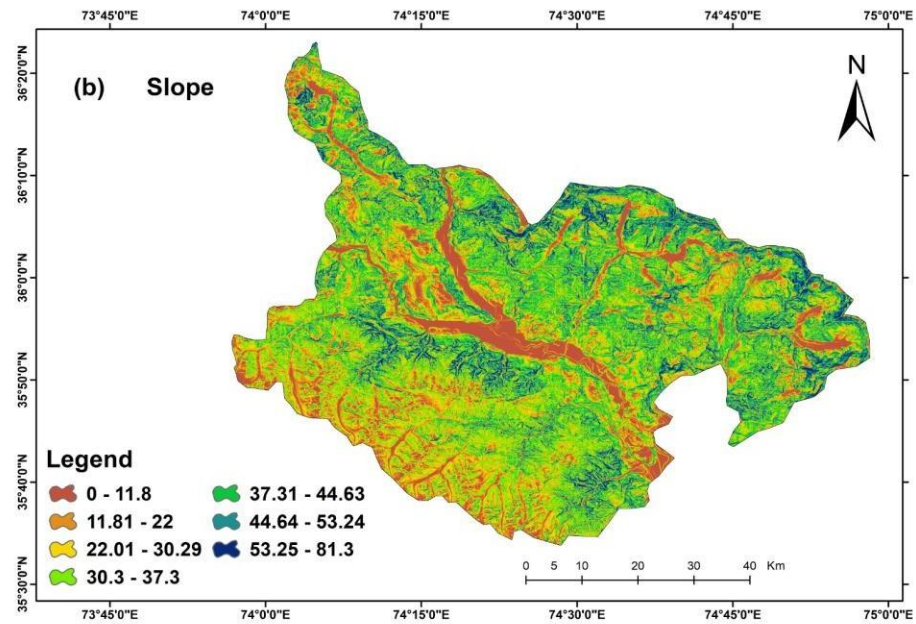


**Figure 3 : b) Slope**

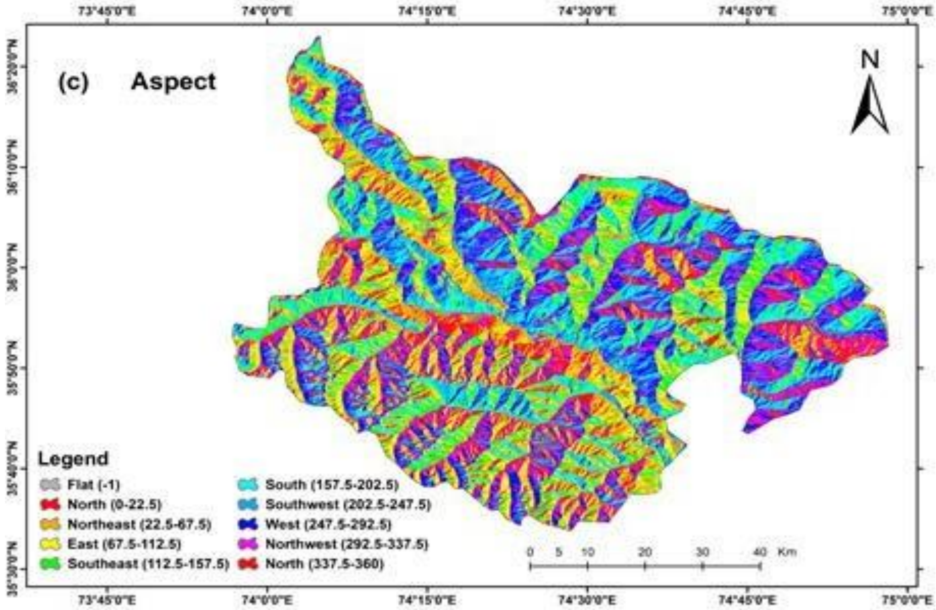


**Figure 3 : c) Aspect**

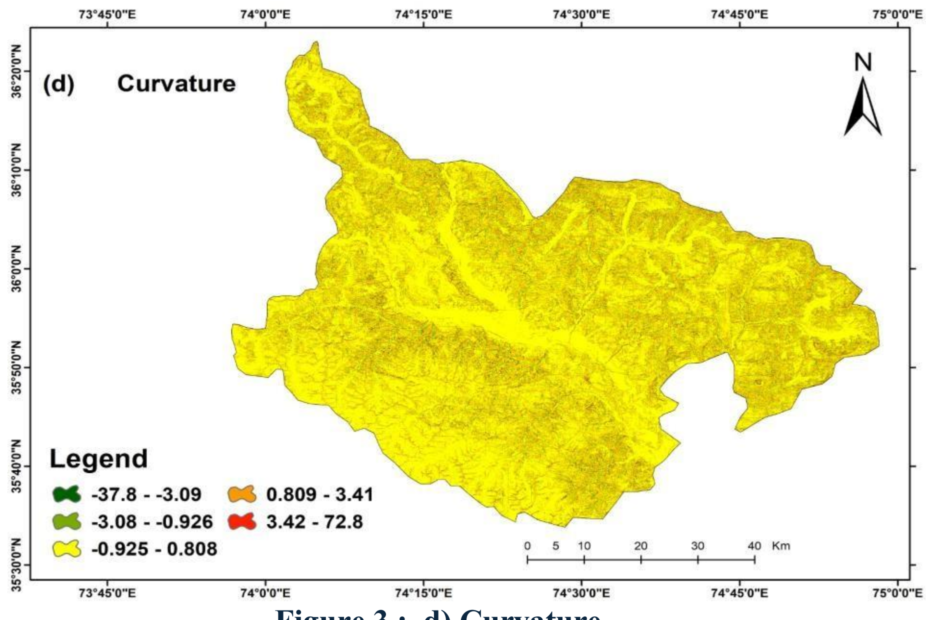


**Figure 3 : d) Curvature**

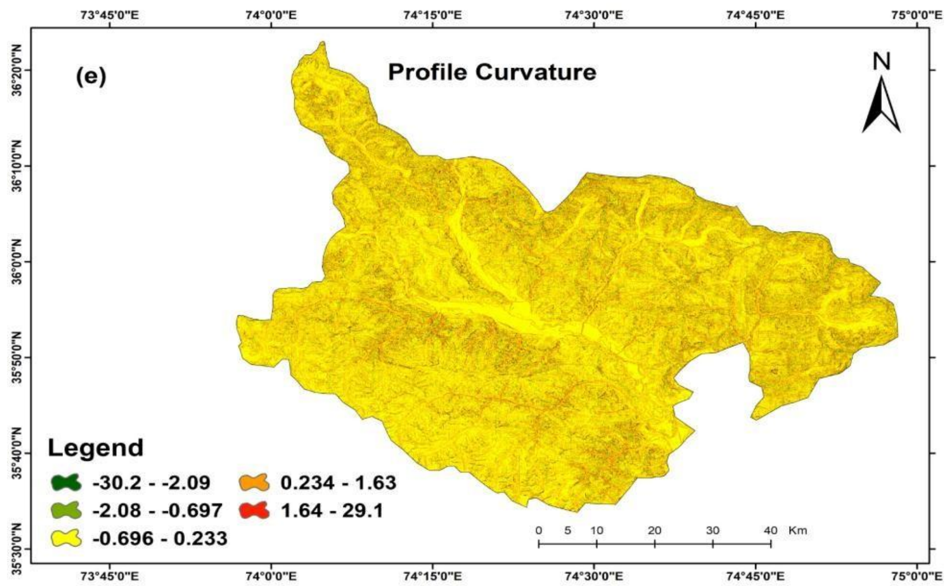


**Figure 3 : e) Profile curvature**

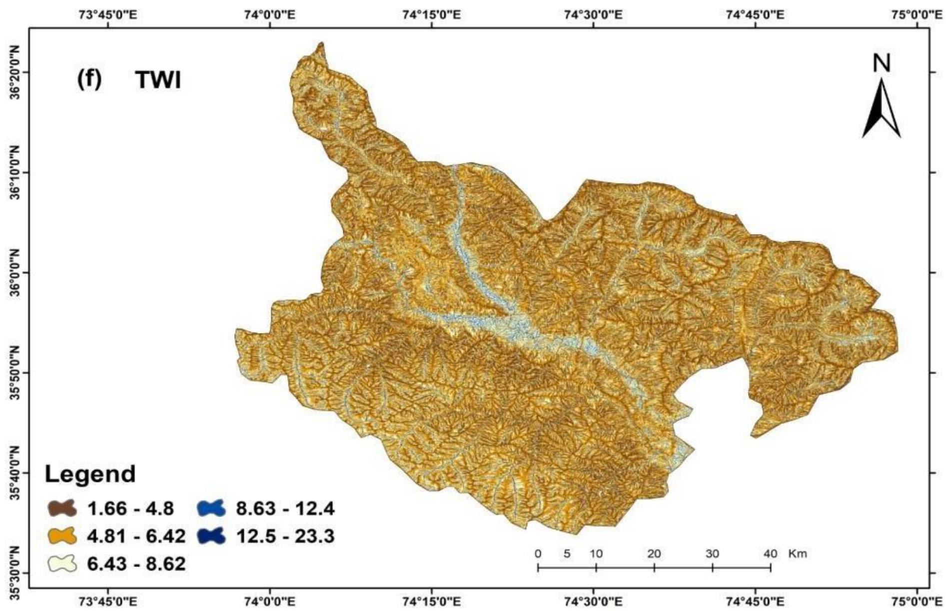


**Figure 3 : f) TWI**

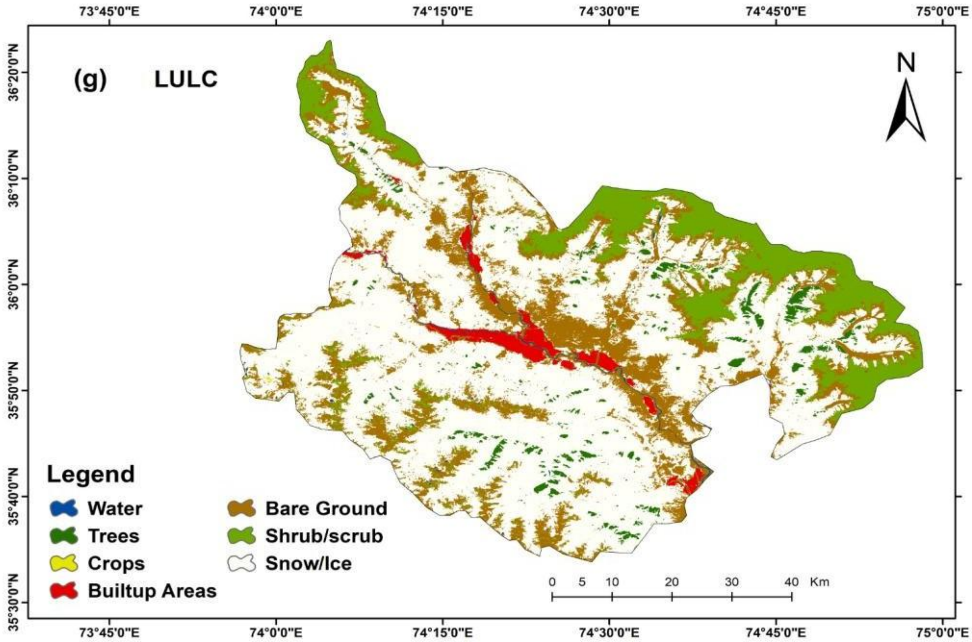


**Figure 3 : g) LULC**

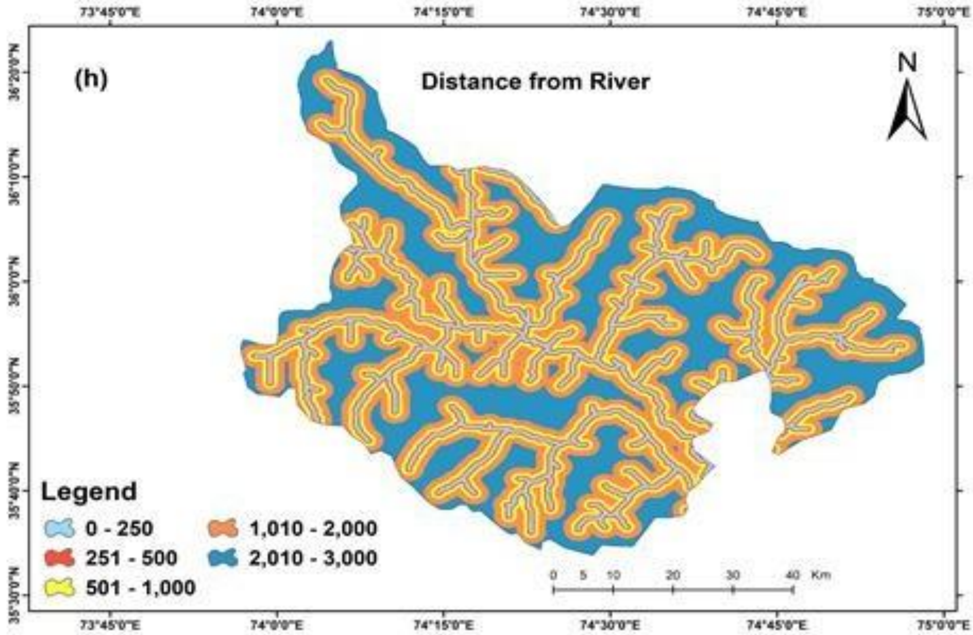


**Figure 3 : h) Distance from River**

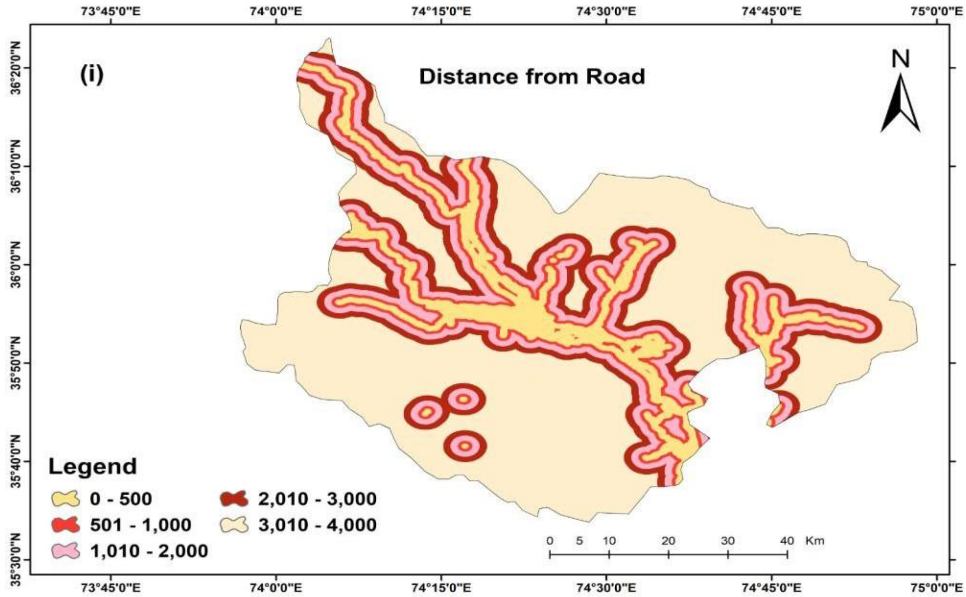


**Figure 3 : i) Distance from Road**

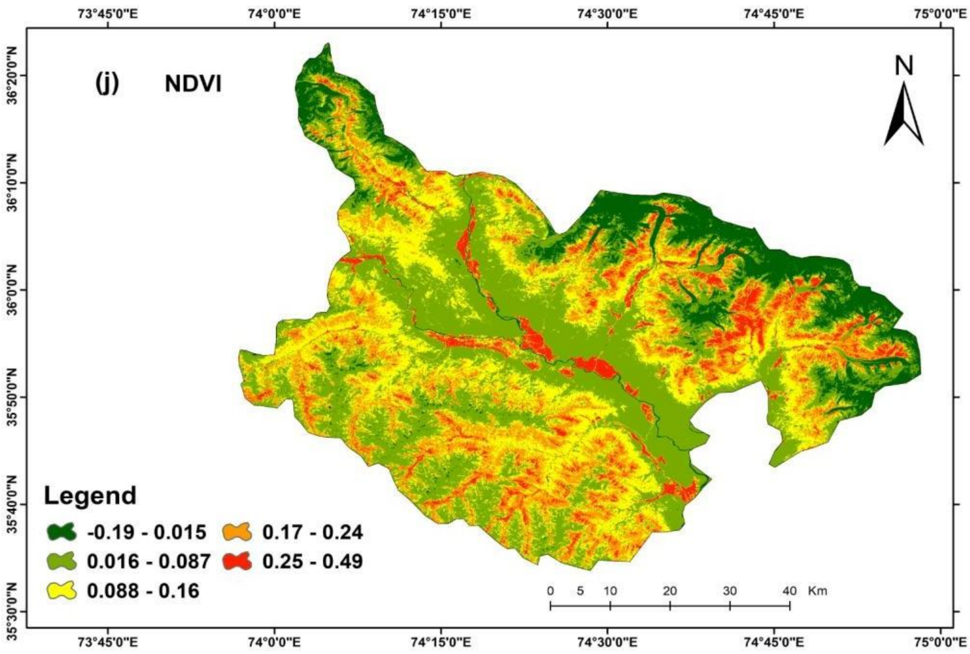


**Figure 3 : j) NDVI**

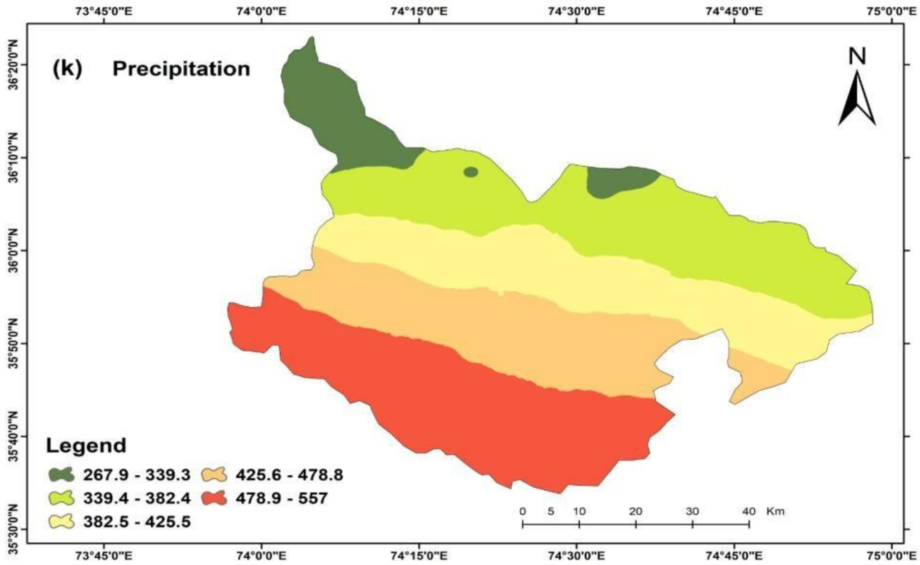


**Figure 3 : k) Precipitation**

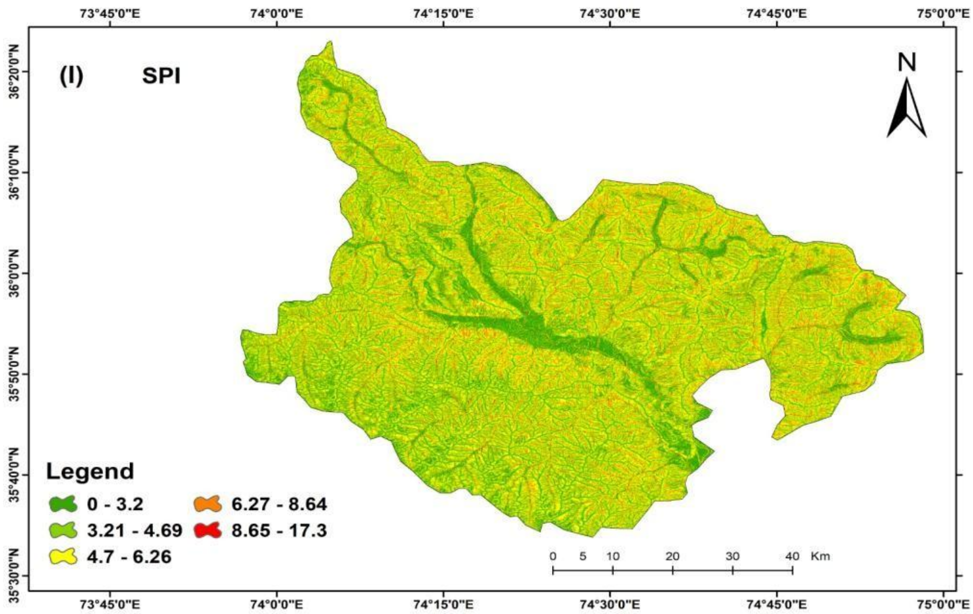


**Figure 3 : l) SPI**

**Figure 3:Landslide Conditioning Factors**

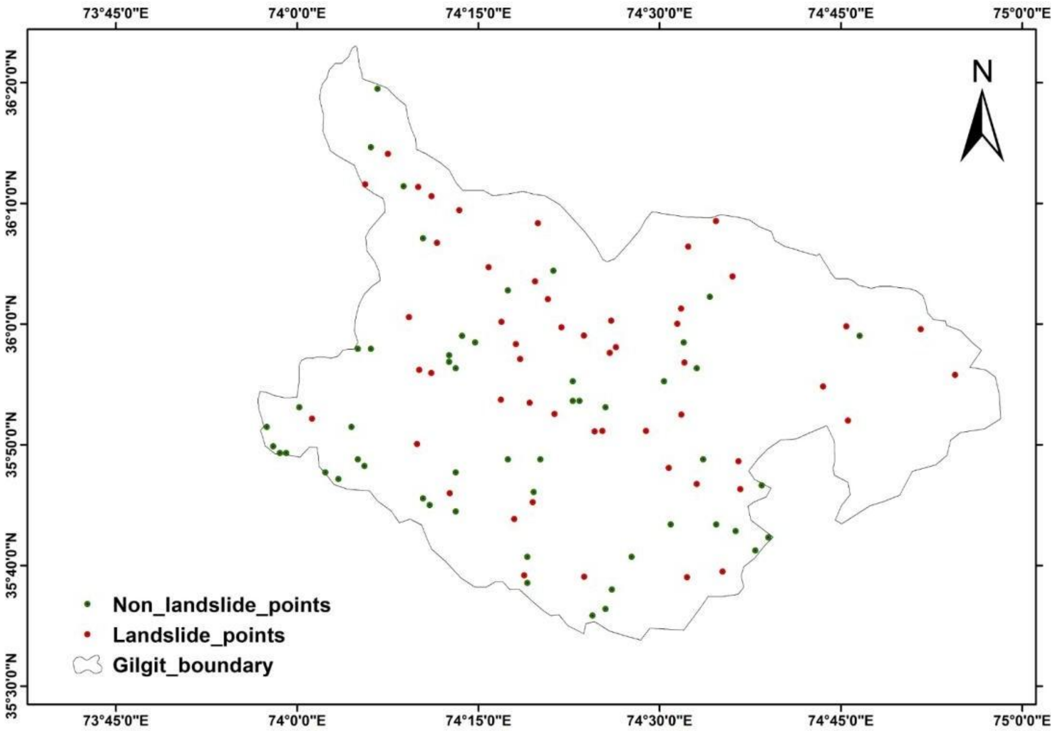


**Figure 4:Spatial Distribution of Landslide and Non-Landslide Inventory Points, Gilgit District (2017–2024)**

# CHAPTER 4

# RESULTS AND DISCUSSION

## 4.1 Statistical Diagnosis of Spatial Multicollinearity

To assess the relationship between the twelve landslide conditioning factors, multicollinearity analysis has been carried out before training the Random Forest model. Multicollinearity is a situation in which the predictor variables are highly correlated, potentially resulting in redundancy and influencing model interpretation (Khaliq et al., 2023; Reichenbach et al., 2018).

All of the conditioning factors were checked for multicollinearity using the Variance Inflation Factor (VIF) and Tolerance (TOL) values. Tolerance is the percent of the variance of a variable that is not accounted for by other predictors, and VIF is a measure of the amount of correlation among variables (Hair et al., 2010). The mathematical expressions are:

$$TOL = 1 - R^2$$

$$VIF = 1 / TOL$$

Where $R^2$ is the coefficient of determination obtained by regressing a variable against all remaining predictors.

The commonly accepted thresholds are a VIF > 5 and a Tolerance < 0.20, suggesting that the variables are problematic with respect to multicollinearity and should be considered for removal (Hair et al., 2010).

The empirical outputs are presented systematically in Table 4.1.

**Table 1: Results of Multicollinearity Analysis**

| No. | Variables | Tolerance (TOL) | VIF |
|---|---|---|---|
| **1** | Elevation | 0.27 | 3.62 |
| **2** | Slope | 0.51 | 1.93 |
| **3** | Aspect | 0.92 | 1.08 |
| **4** | Curvature | 0.21 | 4.65 |
| **5** | Profile Curvature | 0.25 | 3.90 |
| **6** | TWI | 0.26 | 3.75 |
| **7** | SPI | 0.41 | 2.43 |
| **8** | LULC | 0.79 | 1.25 |
| **9** | NDVI | 0.87 | 1.13 |
| **10** | Precipitation | 0.66 | 1.50 |
| **11** | Distance from River | 0.45 | 2.19 |
| **12** | Distance from Road | 0.40 | 2.49 |

Source: Author own computation

The results of the tests shown in Table 4.1 show that all 12 conditioning factors meet the acceptable criterion for multicollinearity. The highest VIF value was observed for Curvature (4.65) and the lowest Tolerance value was 0.21, both of which were within limits. The other variables had VIF values less than 5 and Tolerance values greater than 0.20.

The results show that there is no strong linear dependency among the conditioning factors selected. All these 12 variables were thus retained for further modelling with Random Forest. There is no strong multicollinearity among the factors, which indicates that each factor provides independent information for the landslide susceptibility assessment and ensures the reliability of the developed predictive model (Basharat et al., 2021; Khaliq et al., 2023).

### 4.2 Discussion of Model Performance

Table 4.2 shows the performance of the models, with the Random Forest model showing good performance in the classification of landslide and non-landslide locations in the study area. The Overall Accuracy of the model was 84.4%, which is a good indicator of its predictive capability for landslide susceptibility assessment in the Gilgit District.

The Precision is 0.87, meaning that 87% of the locations that were predicted as landslides were accurately classified. The same applies to the Recall value of 0.81, indicating the model adequately detected 81% of the landslide locations in the testing data set. The F1-Score of 0.84 represents a balance between the precision and recall, which means that the classification performance is stable.

The classification report also confirmed the balanced behaviour of the models for both classes. The non-landslide class obtained an F1 of 0.85 and the landslide class obtained an F1 of 0.84. These results indicate that there was no significant class bias in this model and that it was predicting equally well across the data set.

The results obtained are similar to the previous landslide susceptibility studies carried out in mountainous areas of northern Pakistan. Other studies have found strong performances from Random Forest models because they are able to capture the complex and non-linear relationships between the environmental conditioning factors (Hussain et al., 2021; Khaliq et al., 2023).

**Table 2: Performance metrics of ML model Random Forest**

| Evaluation Metric | Value | Percentage |
|---|---|---|
| Precision | 0.87 | 87.0% |
| Recall | 0.81 | 81.0% |
| F1-Score | 0.84 | 84.0% |
| Overall Accuracy | 0.84 | 84.4% |

Source: Authors own computation

To further investigate the model discrimination ability, Receiver Operating Characteristic (ROC) analysis was used. A ROC is a graph that shows how the True Positive Rate (Sensitivity) increases with the False Positive Rate at various classification thresholds. Model discrimination capability was excellent with an Area Under the Curve (AUC) of 0.94. The AUC > 0.9 is considered as an excellent value and the result indicates that the Random Forest model is a good indicator for the class separation of susceptible and non-susceptible locations and can be confidently used for landslide susceptibility mapping in the Gilgit District.

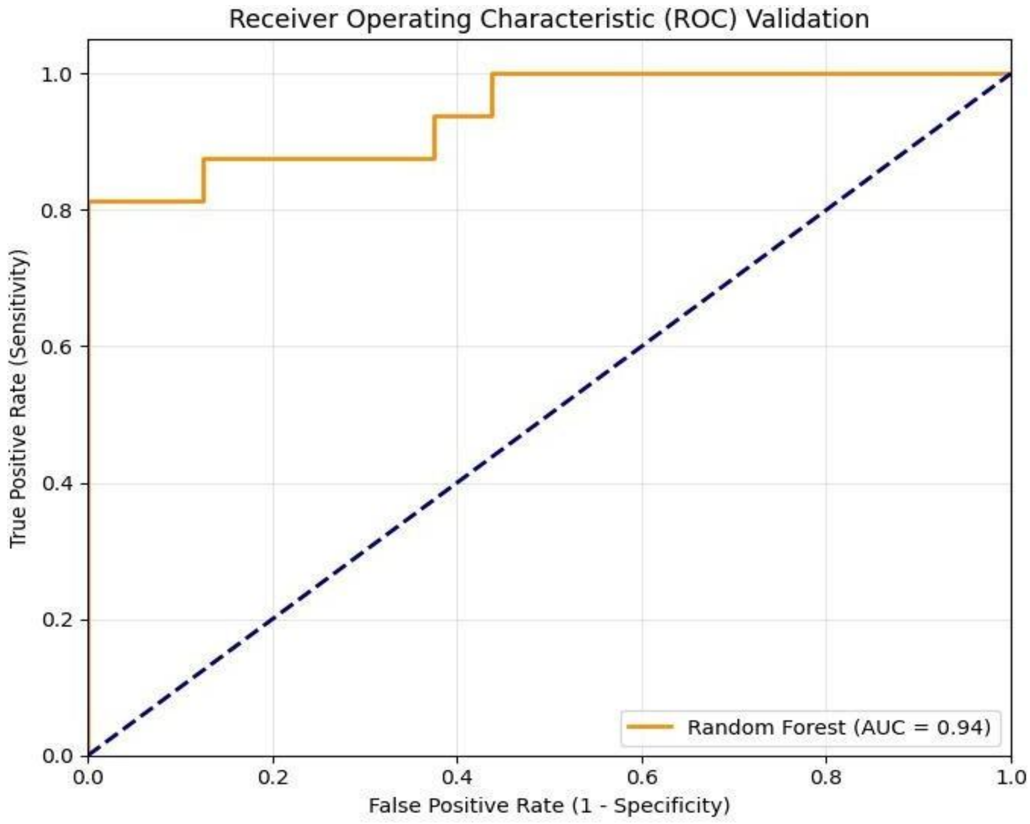


**Figure 5: ROC-AUC Validation Curve of the Random Forest Model (AUC = 0.94), Gilgit District**

### 4.3 Spatial Delineation of Landslide Susceptibility Zones

The twelve conditioning factors were retrieved from the model to investigate the relative importance after checking the predictive performance of the Random Forest model. This importance values were normalized and applied as weights in a weighted overlay analysis using Geographic Information System (GIS) to draw the following Gilgit District final landslide susceptibility map.

All reclassified conditioning factors (Elevation, Slope, Aspect, Curvature, Profile Curvature, TWI, SPI, LULC, NDVI, Mean Annual Precipitation, Distance from Rivers, and Distance from Roads) were combined using the weighted overlay approach.

The final susceptibility index was created in ArcGIS Pro using Raster Calculator to create a continuous landslide susceptibility surface for the entire study area. The susceptibility index obtained was then categorized into five susceptibility classes based on the Natural Breaks (Jenks) classification method (Jenks, 1967) for ease in interpretation and for applications in disaster management. The five classes are: Very Low, Low, Moderate, High, and Very High susceptibility zones.

Spatial susceptibility classes show that the High and Very High susceptibility areas are primarily located along the steep valley slopes, river courses and mountainous areas with high relief and active geomorphological processes. These are well pronounced in the valleys of Gilgit River and Hunza River, where the steep slopes, active erosion and anthropogenic activities raise the slope instability (Alvi et al., 2020; Hussain et al., 2021).

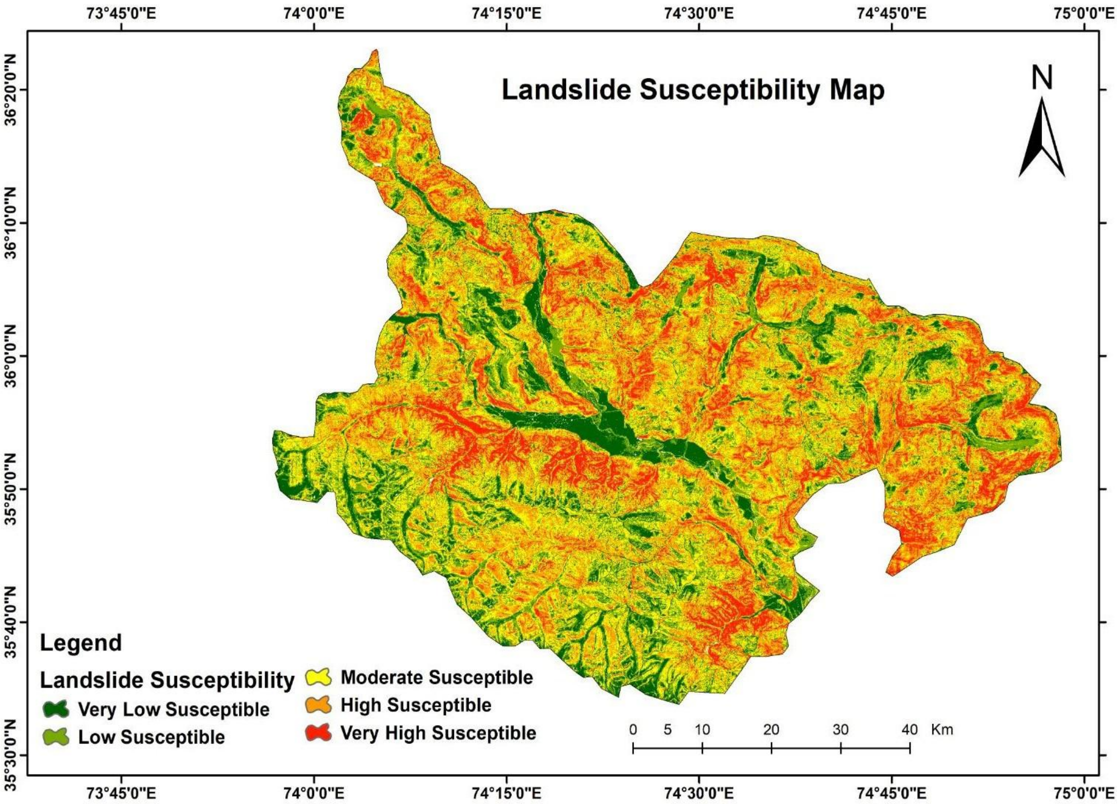


**Figure 6: Final Landslide Susceptibility Map of Gilgit District (2017–2024) Random Forest Model**

### 4.4 Model Interpretability and Factor Contribution Analysis

#### 4.4.1 Random Forest Feature Importance Analysis

The importance of the variables generated by the Random Forest model is shown in Figure 5. The results showed that Slope had the highest influence on the occurrence of landslides in the Gilgit District followed by NDVI, Distance from Roads, Distance from Rivers, TWI, and Elevation.

The importance of controlling slope instability by gravitational forces is again confirmed by the dominance of Slope. In the same way, proximity to roads and rivers indicates the impact of anthropogenic disturbances of the slope and river-bank erosion, respectively, while NDVI indicates density of vegetation and root cohesion. Profile Curvature and Curvature were among minor factors affecting the model prediction. The feature importance analysis gives an overall ranking of the importance of the predictor but does not account for if a high or low value of a predictor will make the landslide more or less susceptible.

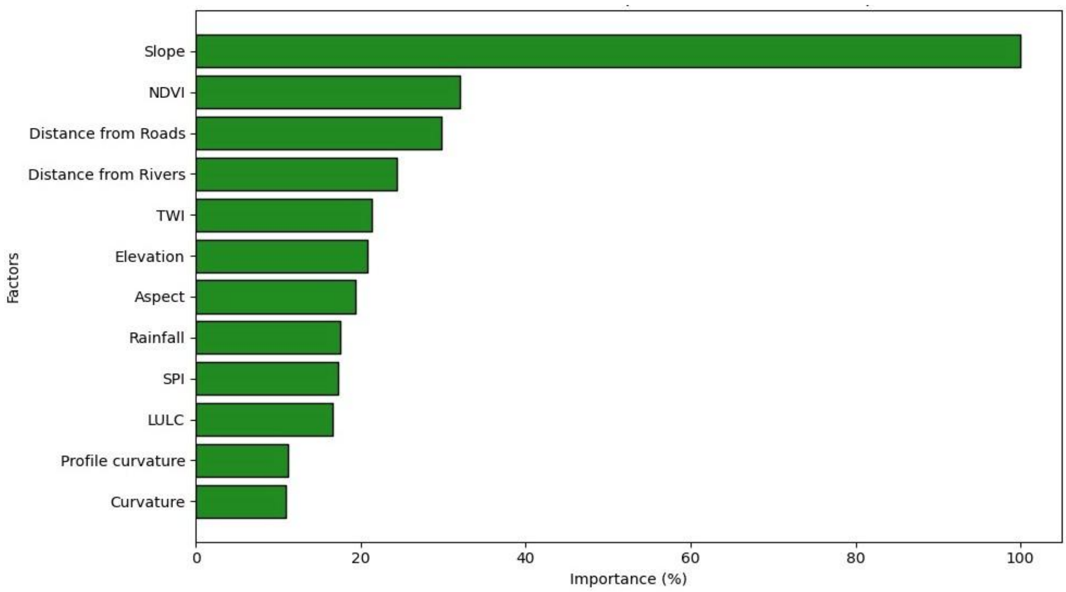


**Figure 7: Random Forest Feature Importance**

### 4.4.2 SHAP-Based Explainable AI Interpretation

Additive exPlanations (SHAP) were used to enhance the transparency of the model and to deal with the black box approach to machine learning. SHAP provides the individual contribution of each conditioning variable to increase or decrease the landslide susceptibility predictions.

The SHAP summary plot revealed Slope as the most important factor affecting the occurrence of landslides in the Gilgit District. Positive SHAP values are correlated with high slopes, which mean that the probability of landslides is higher in that area, while negative SHAP values correlate to low slope areas, which mean that the probability of landslides is lower in that area.

The second most influential factor was found to be NDI. The areas with low vegetation cover had positive contributions to landslide occurrence, whereas the high vegetation cover areas tended to decrease the susceptibility. Likewise, LULC had a strong influence; unstable classes such as barren and sparsely vegetated are more susceptible to slope failure than stable classes.

Susceptibility was also significantly affected by Distance from Roads and Distance from Rivers. Slides were more likely to occur in areas near transportation corridors and active river channels as a result of road-cutting activities and river bank erosion processes.

Rainfall was a positive influencer of susceptibility, thus reinforcing its role as an important triggering factor. The comparatively smaller effects, but significant in creating local terrain and hydrological conditions, were observed in the variables: Elevation, TWI, SPI, Aspect, Curvature and Profile Curvature.

In general, the SHAP analysis indicated that the topographic conditions, vegetation characteristics, land-cover patterns and proximity to roads and rivers are the major influences on landslide occurrence in the Gilgit District.

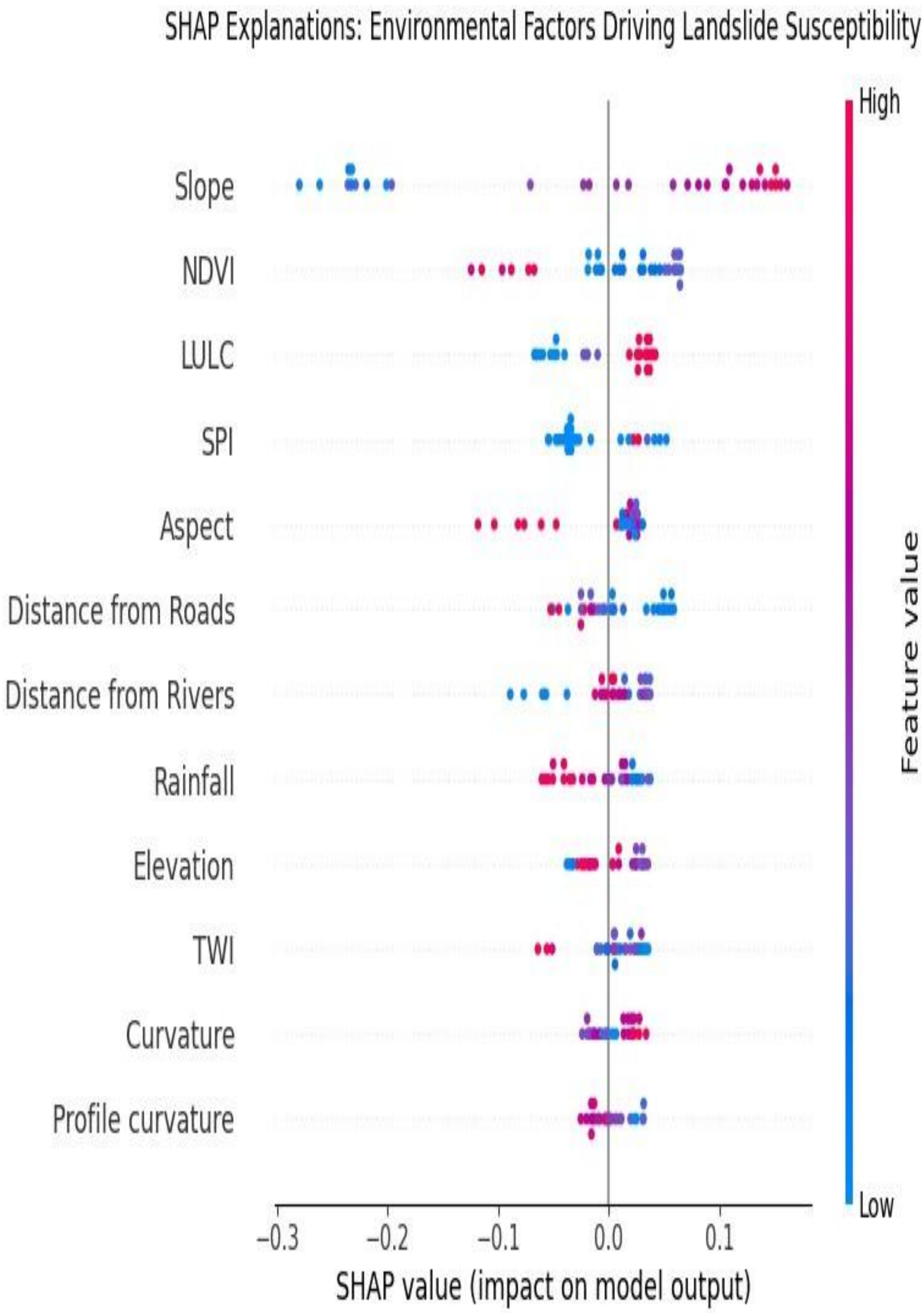


**Figure 8: SHAP Summary Plot**

# CHAPTER 5

# CONCLUSION AND RECOMMENDATIONS

## 5.1 Conclusion

A Landslide Susceptibility Mapping (LSM) framework is successfully developed for the Gilgit District, Gilgit-Baltistan, in this study using Remote Sensing, GIS, Machine Learning and Explainable Artificial Intelligence (XAI) techniques. The research was carried out over a seven-year period (2017–2024) to characterize the landslide prone areas, and to assess the relative contribution of anthropogenic and environmental factors to slope instability. The inventory points of landslides were done by visual interpretation with Google Earth Pro and cross-checked with available disaster records, and non-landslides points were created in ArcGIS for supervised machine learning modelling.

A total of twelve landslide conditioning factors, such as Elevation, Slope, Aspect, Curvature, Profile Curvature, TWI, SPI, LULC, NDVI, Rainfall, Distance from Rivers, Distance from Roads were prepared and standardized for model development. Multicollinearity analysis was conducted before the training of the models using Variance Inflation Factor (VIF) and Tolerance (TOL). The results showed that all variables were within acceptable limits, which suggested that there were no serious multicollinearity issues and hence they are suitable for the model implementation (Hair et al., 2010; Khaliq et al., 2023).

The model built with Random Forest (RF) had the best prediction performance with an Overall Accuracy of 84.4%, a Precision of 87.0%, a Recall of 81.0%, an F1-Score of 84.0% and an AUC value of 0.94. The results showed that the model was capable of differentiating between landslide prone areas and stable areas, and yielded useful susceptibility maps throughout the study area (Basharat et al., 2021; Merghadi et al., 2020).

To understand the model, the variation that each conditioning factor contributed to the model was analyzed by using SHapley Additive exPlanations (SHAP). The SHAP analysis showed that the most important factor affecting the occurrence of landslides was Slope, followed by NDVI, Distance from Roads, Distance from Rivers, TWI, and Elevation. From the results it can be concluded that steepness of slopes, low vegetation cover and road or river channels nearby greatly influence the susceptibility to landslides in the district (Lundberg & Lee, 2017; Khaliq et al., 2023; Hussain et al., 2021).

Based on the final landslide susceptibility map, the study area was divided into five different landslide susceptibility zones of Very Low to Very High. High and Very High susceptibility areas were predominantly located in steep valleys and river corridors and transportation routes, especially along the Karakoram Highway. Thus, the susceptibility map created is a useful tool for decision making in the field of disaster risk reduction, infrastructure development and sustainable land use in Gilgit District and other mountainous areas of Northern Pakistan (Saira and Miandad, 2023; Hussain et al., 2021).

## 5.2 Recommendations for Strategic Disaster Risk Management

Based on the susceptibility patterns derived by the Random Forest model and the factor importance analysis derived from SHAP, the following recommendations are proposed to support the landslide risk reduction and sustainable development in the Gilgit District.

### 5.2.1 Integration of Landslide Susceptibility Maps into Spatial Planning

The final Landslide Susceptibility Map should be integrated into regional planning and development activities by the Gilgit-Baltistan Disaster Management Authority (GBDMA), local government departments, and planning agencies (2024). The High and Very High susceptibility areas should be taken into consideration with precaution prior to the development of additional residential, commercial structures, and public infrastructure. Such integration can be used to minimize future exposure to landslide hazards and to aid in informed land-use planning. This is in line with the disaster risk reduction principles stated in the Sendai Framework for Disaster Risk Reduction 2015-2030 (UNISDR, 2015)002E

### 5.2.2 Promotion of Bio-Engineering and Vegetation-Based Slope Stabilization

Vegetation cover was identified as a significant factor in the SHAP analysis that decreases landslide susceptibility. In this regard, bio-engineering measures like afforestation and planting of indigenous deep rooted vegetation would be encouraged on vulnerable slopes (Piacentini et al., 2012; Khaliq et al., 2023). The addition of vegetation can increase slope stability, soil cohesion and decrease surface erosion, especially in areas that are known to be highly susceptible to landslides.

### 5.2.3 Improvement of Engineering Practices along Transportation Corridors

Considering the importance of the proximity of road on landslide occurrence, road construction and expansion projects could be accompanied by detailed geotechnical assessments prior to the implementation of such projects (Saira & Miandad, 2023; Iqbal et al., 2020). To prevent slope instability hazards, it is necessary to implement appropriate mitigation measures along vulnerable sections of the Karakoram highway and the related road network such as retaining structures, slope reinforcement techniques, drainage management systems and rockfall protection barriers.

### 5.2.4 Development of Localized Landslide Early Warning Systems

Rainfall was determined to be one of the factors that affects landsliding, therefore, developing localized landslide monitoring and early warning systems is recommended (Crozier, 2010; Moazzam et al., 2022). In landslide prone areas, Pakistan Meteorological Department (PMD) needs to increase rainfall monitoring by installing more automated weather stations and real-time data collection systems (PMD, 2024). The susceptibility map created in this study can be used to pinpoint areas to be monitored and assessed. The incorporation of susceptibility data with rainfall observations could enhance hazard preparedness and enable effective risk communication to vulnerable populations in a timely fashion. This is in line with the disaster risk reduction strategies of UNISDR (2015).

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